



Final Exam (Take-Home-Task)

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Date Completed: 19 December 2025

Laboratory Exercise Title: Vending Machine Controller (VMC)

Target Course Outcomes:

CO1: Create descriptions of digital hardware components such as in combinational and sequential circuits using synthesizable Verilog HDL constructs.

CO2: Verify the functionality of HDL-based components through design verification tools.

CO3: Design HDL-based digital systems following the design flow for FPGA implementation.

Final Exam (Take-Home-Task): Vending Machine Controller (VMC)

Objective

In this exercise, a Vending Machine Controller (VMC) is designed in Verilog HDL to simulate a real-world automated retail system. The design features a Finite State Machine (FSM) with seven distinct states controlling the selection, payment, dispensing, and change-return processes. The system supports three distinct items with prices set at 3, 5, and 12 Pesos, and accepts coin inputs of 1, 5, and 10 Pesos. It utilizes a greedy algorithm to calculate and dispense change efficiently. The design includes a top-level wrapper (VMC_DE10) that integrates a clock divider for human-perceptible timing, debouncing logic, and 7-segment display drivers to visualize Cost, Paid Amount, and Change Due in real-time on the DE10-Lite FPGA board.

Entity Diagram and I/O Ports

- **Inputs:**
 - CLOCK → Negative-edge triggered system clock (Derived from MAX10_CLK1_50 via ClockDivider)
 - nRESET → Asynchronous active-low reset (Resets state to IDLE and clears registers)
 - START → Synchronous active-high input to initiate a transaction
 - OK → Synchronous active-high input to confirm selection or payment
 - CANCEL → Active-high input to abort transaction and trigger a full refund
 - SELECT → Synchronous active-high input to cycle through available items
 - COIN_1 / COIN_5 / COIN_10 → Coin mechanism inputs representing 1, 5, and 10 Peso insertions
- **Outputs:**
 - ITEM[2:0] → 3-bit one-hot encoded output driving LEDs to indicate the currently selected product
 - DISPENSE → Register-driven output signal to trigger the product dispensing actuator
 - C1 / C5 / C10 → Change actuator outputs (active high) for returning 1, 5, and 10 Peso coins
 - DBG_COST[7:0] → 8-bit debug bus outputting the current item price
 - DBG_PAID[7:0] → 8-bit debug bus outputting the current accumulated balance
 - DBG_CHANGE[7:0] → 8-bit debug bus outputting the calculated change

Modeling Method



Modeling Type: Behavioral modeling using a mix of sequential logic for state transitions/register updates and combinational logic for 7-segment decoding and LED indicators. The FSM utilizes a case statement to manage transitions between S_IDLE, S_SELECT, S_PAY, S_DISPENSE_ITEM, S_CALC_CHANGE, S_REFUND, and S_OUT_CHANGE.

Verification: The design logic is verified through the VMC_DE10 top-level module, which maps internal signals to the DE10-Lite hardware.

- **Visual Output:** Hexadecimal values for Paid, Change, and Cost are decoded onto HEX0–HEX5.
- **Actuators:** LEDR signals represent the dispensing and coin return mechanisms.

Language: Verilog HDL

File Name: VMC.v

Project Name: VMC

Family: Max 10

Device: 10M50DAF484C7G

Design Flow Summary (Quartus Prime)

After successful synthesis:

Logic Elements Used: 256

Pins Used: 69

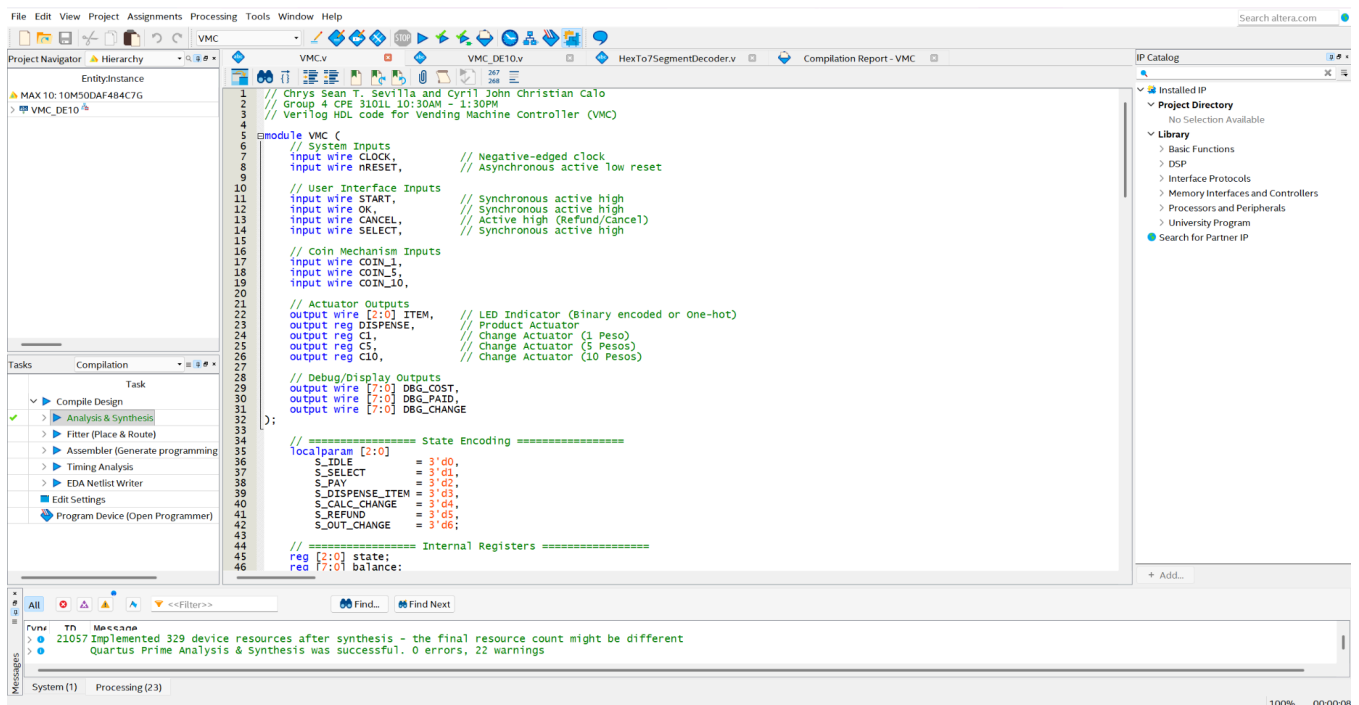


Figure 1. Verilog HDL Design Entry

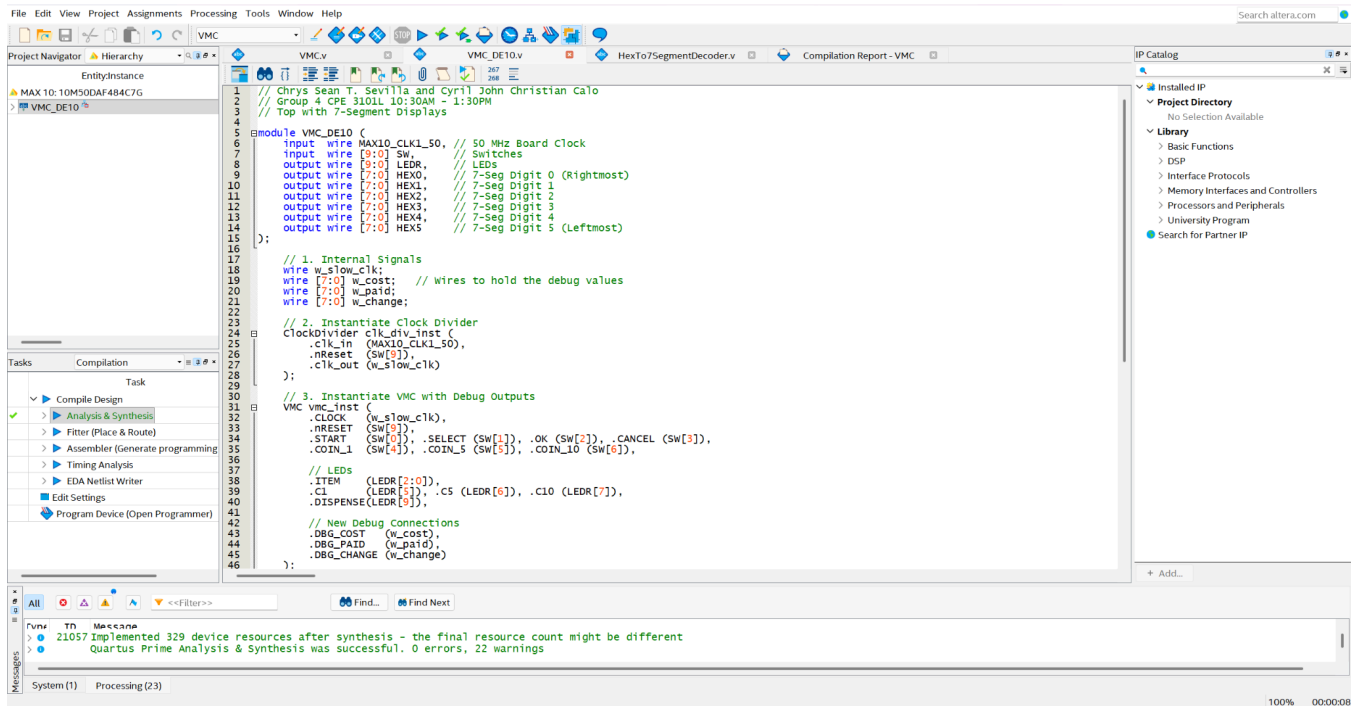


Figure 2. Verilog HDL Design Entry

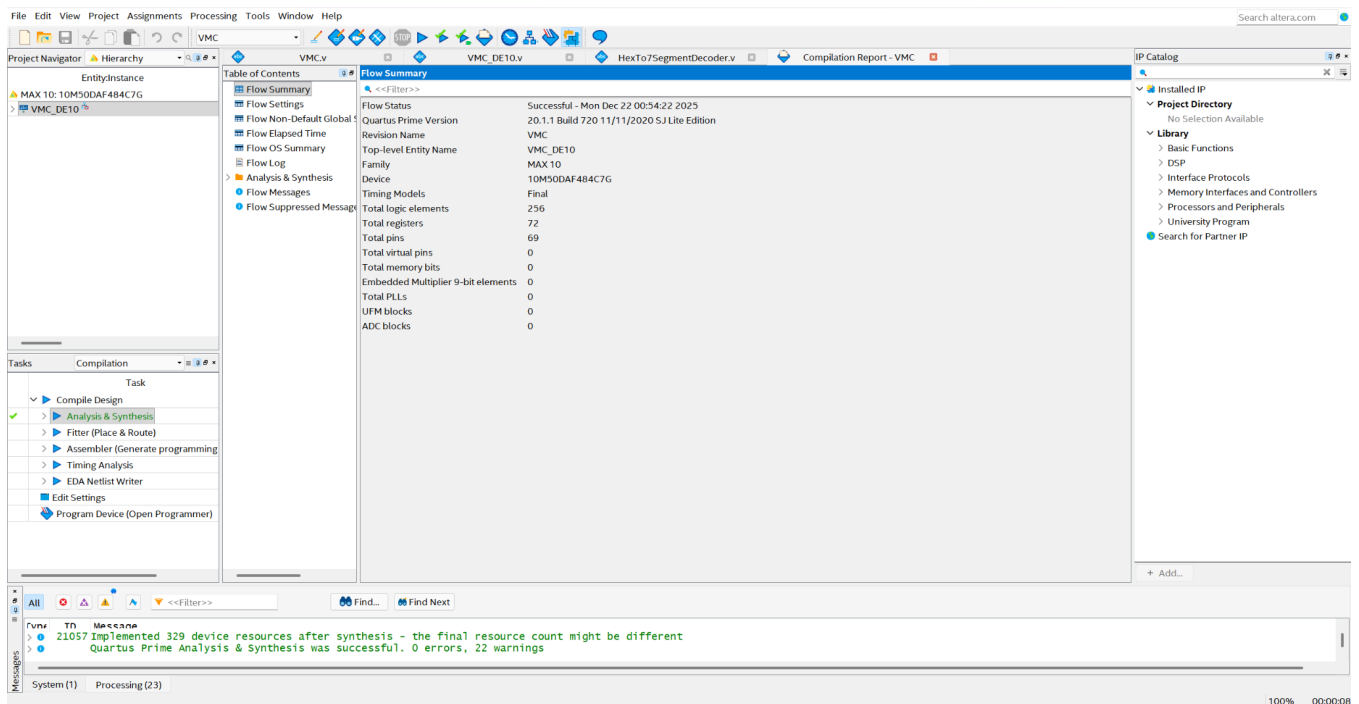


Figure 3. Compilation Report for the Flow Summary

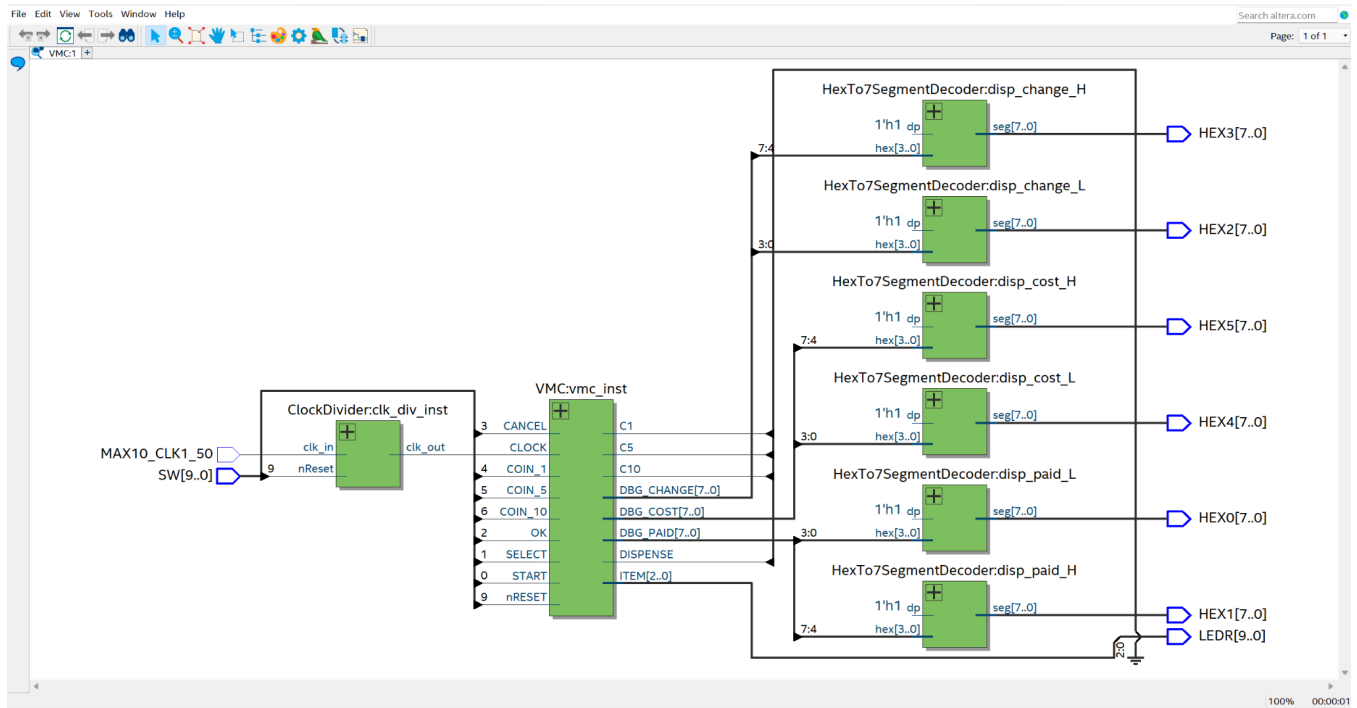


Figure 4. Schematic Diagram of Synthesized Circuit

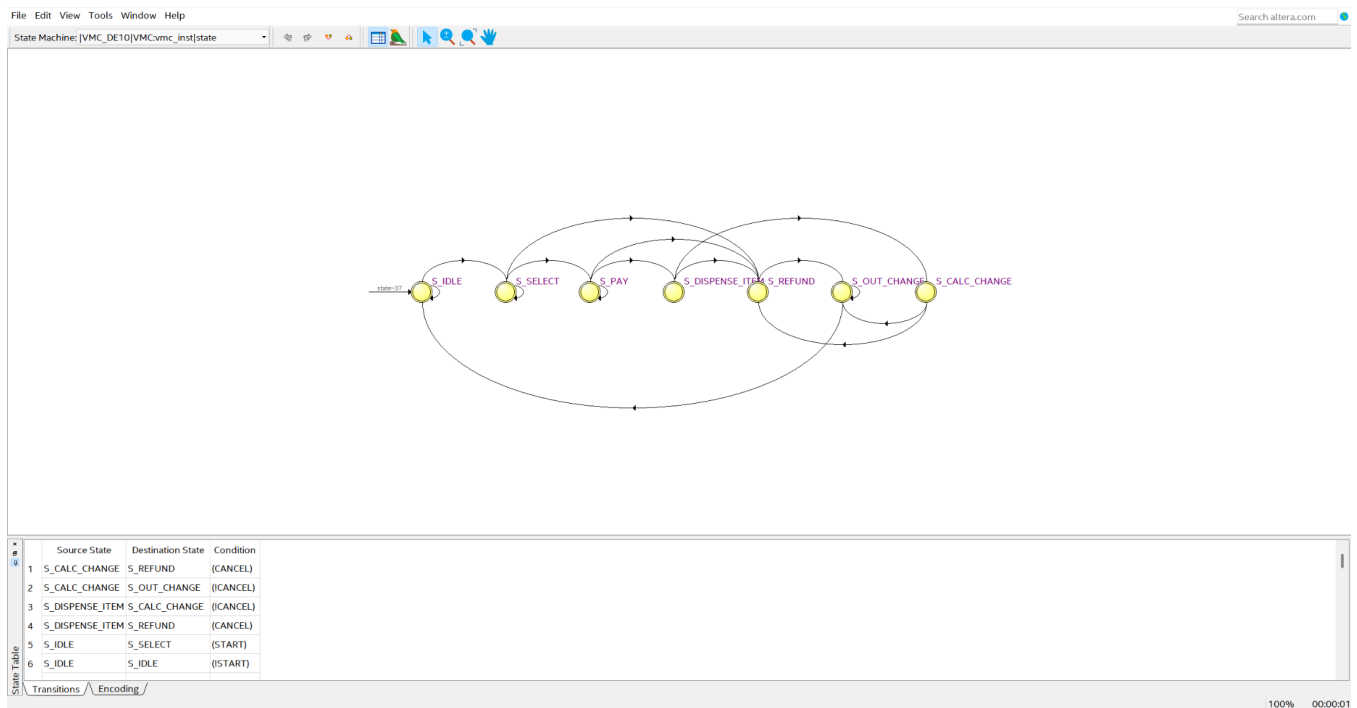


Figure 5. State Machine with State Transition Table



```
1 // Chrys Sean T. Sevilla and Cyril John Christian Calo
2 // Group 4 CPE 3101L 10:30AM - 1:30PM
3 // Verilog HDL code for Vending Machine Controller Testbench (tb_VMC)
4
5 `timescale 1ns/1ps
6
7 module tb_VMC;
8
9 // 1. Signal Declarations
10 // Inputs (Regs)
11 reg CLOCK = 0;
12 reg nRESET = 0;
13 reg START = 0;
14 reg OK = 0;
15 reg CANCEL = 0;
16 reg SELECT = 0;
17 reg COIN_1 = 0;
18 reg COIN_5 = 0;
19 reg COIN_10 = 0;
20
21 // Outputs (Wires)
22 wire [2:0] ITEM;
23 wire DISPENSE, C1, C5, C10;
24
25 // Debug Outputs
26 wire [7:0] DBG_COST;
27 wire [7:0] DBG_PAID;
28 wire [7:0] DBG_CHANGE;
29
30 // 2. Unit Under Test (UUT) Instantiation
31 VMC dut (
32     .CLOCK(CLOCK),
33     .nRESET(nRESET),
34     .START(START),
35     .OK(OK),
36     .CANCEL(CANCEL),
37     .SELECT(SELECT),
38     .COIN_1(COIN_1),
39     .COIN_5(COIN_5),
40     .COIN_10(COIN_10),
41     .ITEM(ITEM),
42     .DISPENSE(DISPENSE)
43 );
44
45 endmodule
```

Messages

Run: 18236 Number of processors has not been specified which may cause overloading on shared machines. Set the global assignment NUM_PARALLEL_PROCESSORS in your QSF to an appropriate value for best performance.

Quartus Prime Netlist Viewers Preprocess was successful. 0 errors, 1 warning

Figure 6. Verilog Testbench File

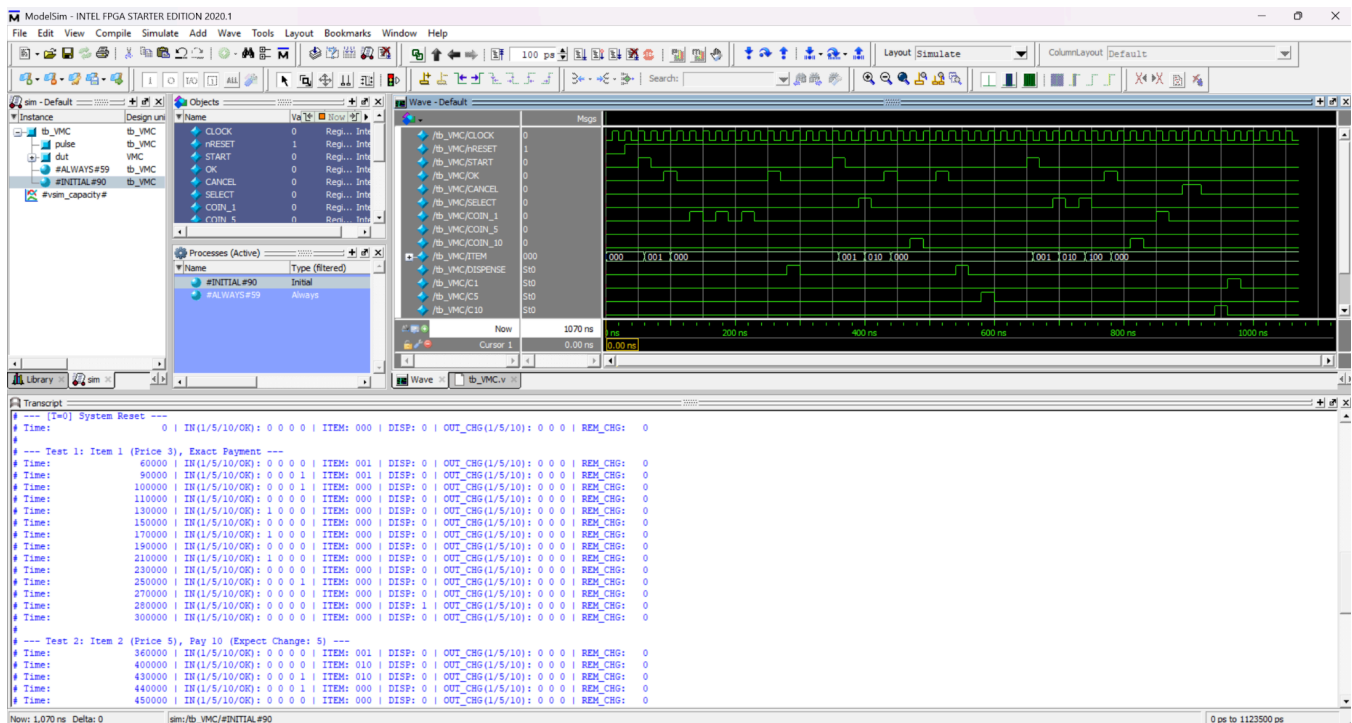


Figure 7. Testbench Waveform with your annotations

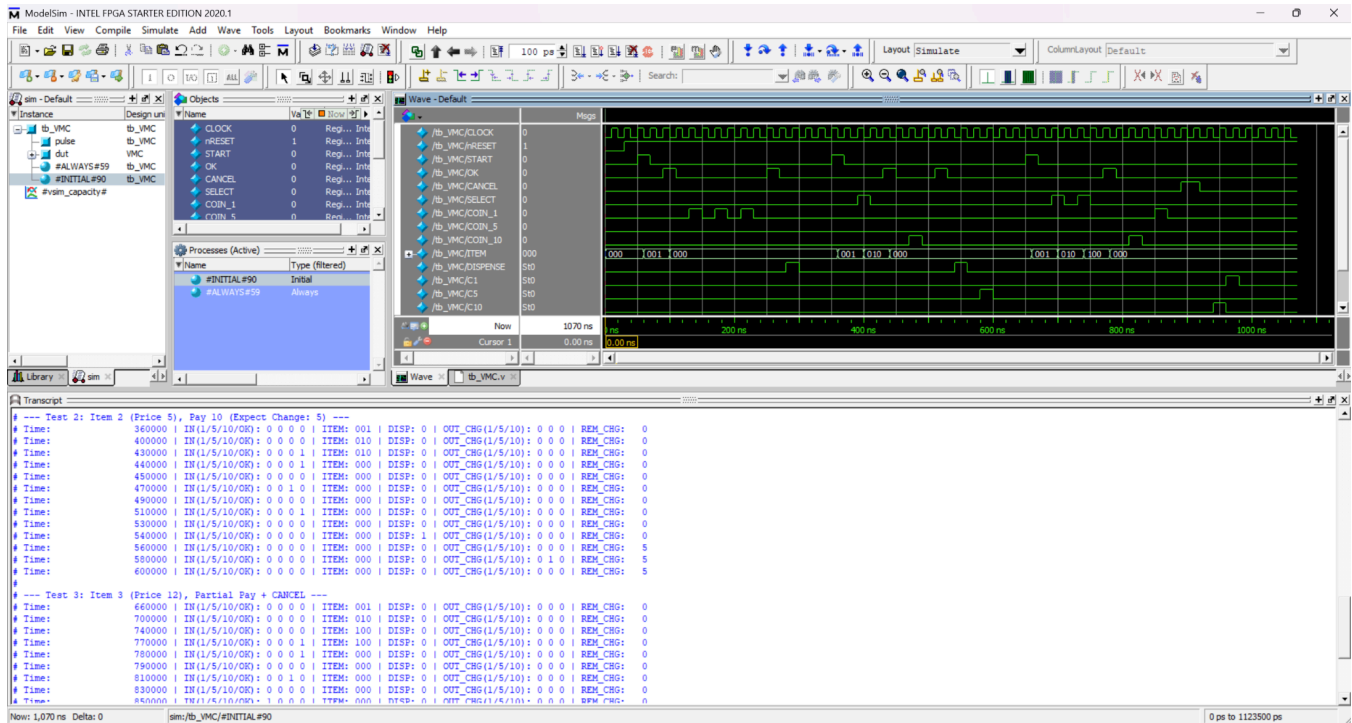


Figure 8. Testbench Waveform with your annotations

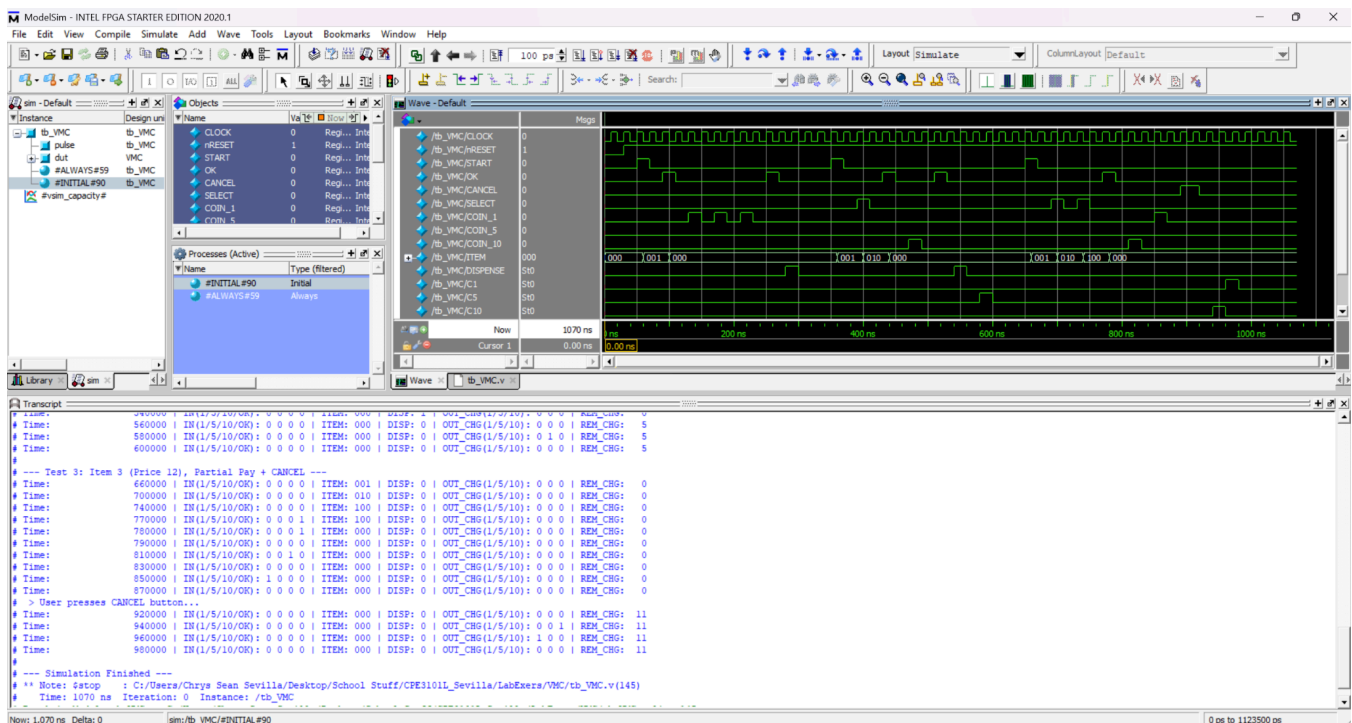


Figure 9. Testbench Waveform with your annotations



Pin Assignments

Switches (Inputs)

SW[0]: START (Initiate Transaction)

SW[1]: SELECT (Cycle through Items)

SW[2]: OK (Confirm Selection/Payment)

SW[3]: CANCEL (Refund/Abort)

SW[4]: Insert 1 Peso Coin

SW[5]: Insert 5 Peso Coin

SW[6]: Insert 10 Peso Coin

SW[7]: Unused

SW[8]: Unused

SW[9]: nRESET (System Reset, Active Low)

LEDs (Outputs)

LEDR[0]: Item 1 Indicator

LEDR[1]: Item 2 Indicator

LEDR[2]: Item 3 Indicator

LEDR[3]: Unused (OFF)

LEDR[4]: Unused (OFF)

LEDR[5]: Change Dispensed (1 Peso Actuator)

LEDR[6]: Change Dispensed (5 Peso Actuator)

LEDR[7]: Change Dispensed (10 Peso Actuator)

LEDR[8]: Unused (OFF)

LEDR[9]: DISPENSE (Item Release Actuator)

7-Segment Displays (Hex Output)



HEX5 - HEX4: COST (Current Item Price)

HEX3 - HEX2: CHANGE (Calculated Change)

HEX1 - HEX0: PAID (Current Balance Inserted)

